



Name: _____

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Learning Objectives

- Evaluate limits by direct substitution, factoring, and rationalizing
- Apply the Squeeze Theorem and recognize one-sided limits
- Evaluate trigonometric and limit-at-infinity problems
- Understand the definition of the derivative as a limit

Show all steps. For 0/0 or ∞/∞ forms, state the technique used before simplifying.

1. Evaluate the limit by direct substitution.

$$\lim_{x \rightarrow 3} (2x^2 - 5x + 1)$$

Answer: _____

2. Evaluate the limit by factoring.

$$\lim_{x \rightarrow 2} \frac{x^2 - 4}{x - 2}$$

Answer: _____

3. Evaluate the limit by factoring.

$$\lim_{x \rightarrow -3} \frac{x^2 + 5x + 6}{x + 3}$$

Answer: _____

4. Evaluate the limit by rationalizing the numerator.

$$\lim_{x \rightarrow 0} \frac{\sqrt{x+9} - 3}{x}$$

Answer: _____

5. Evaluate the trigonometric limit.

$$\lim_{x \rightarrow 0} \frac{\sin(5x)}{x}$$

Answer: _____

6. Evaluate the limit at infinity.

$$\lim_{x \rightarrow \infty} \frac{3x^2 - 1}{x^2 + 5}$$

Answer: _____



7. Evaluate the one-sided limit.

$$\lim_{x \rightarrow 2^-} \frac{x-2}{|x-2|}$$

Answer: _____

8. Determine if the limit exists. Evaluate both one-sided limits.

$$\lim_{x \rightarrow 1^\pm} f(x), \quad f(x) = x^2 \text{ if } x < 1, \quad f(x) = 2x - 1 \text{ if } x \geq 1$$

Answer: _____

9. Evaluate the limit involving a difference quotient.

$$\lim_{h \rightarrow 0} \frac{(x+h)^2 - x^2}{h}$$

Answer: _____

10. Apply the Squeeze Theorem. Given $-x^2 \leq f(x) \leq x^2$ near $x = 0$, find the limit.

$$\lim_{x \rightarrow 0} f(x) \text{ given } -x^2 \leq f(x) \leq x^2$$

Answer: _____





Problems 4 and 5 introduce rationalization and trig limits ($\sin x/x$) — emphasize the standard result $\lim \sin(u)/u = 1$.
Problem 9 connects limits to the derivative definition.

Solutions

1. Evaluate the limit by direct substitution.

$$\lim_{x \rightarrow 3} (2x^2 - 5x + 1)$$

→ Substitute $x = 3$ directly: $2(9) - 5(3) + 1$.

→ $= 18 - 15 + 1 = 4$.

Answer: $2(3)^2 - 5(3) + 1 = 4$

2. Evaluate the limit by factoring.

$$\lim_{x \rightarrow 2} \frac{x^2 - 4}{x - 2}$$

→ Factor numerator: $x^2 - 4 = (x-2)(x+2)$.

→ Cancel $(x-2)$: limit becomes $\lim(x+2)$ as $x \rightarrow 2$.

→ $= 2 + 2 = 4$.

Answer: $\lim_{x \rightarrow 2} (x+2) = 4$

3. Evaluate the limit by factoring.

$$\lim_{x \rightarrow -3} \frac{x^2 + 5x + 6}{x + 3}$$

→ Factor numerator: $x^2 + 5x + 6 = (x+3)(x+2)$.

→ Cancel $(x+3)$: limit becomes $\lim(x+2)$ as $x \rightarrow -3$.

→ $= -3 + 2 = -1$.

Answer: $\lim_{x \rightarrow -3} (x+2) = -1$

4. Evaluate the limit by rationalizing the numerator.

$$\lim_{x \rightarrow 0} \frac{\sqrt{x+9} - 3}{x}$$

→ Multiply by conjugate $(\sqrt{x+9}+3)/(\sqrt{x+9}+3)$.

→ Numerator becomes $(x+9)-9 = x$.

→ Simplify: $x / [x(\sqrt{x+9}+3)] = 1/(\sqrt{x+9}+3)$.

→ As $x \rightarrow 0$: $1/(\sqrt{9}+3) = 1/6$.

Answer: $\frac{1}{6}$



5. Evaluate the trigonometric limit.

$$\lim_{x \rightarrow 0} \frac{\sin(5x)}{x}$$

→ Rewrite: $\sin(5x)/x = 5 \cdot \sin(5x)/(5x)$.

→ Let $u = 5x$. As $x \rightarrow 0$, $u \rightarrow 0$.

→ Use the standard limit: $\lim(\sin u / u) = 1$ as $u \rightarrow 0$.

→ Result: $5 \cdot 1 = 5$.

Answer: $5 \cdot \lim_{u \rightarrow 0} \frac{\sin u}{u} = 5$

6. Evaluate the limit at infinity.

$$\lim_{x \rightarrow \infty} \frac{3x^2 - 1}{x^2 + 5}$$

→ Divide numerator and denominator by x^2 (highest power).

→ $= (3 - 1/x^2) / (1 + 5/x^2)$.

→ As $x \rightarrow \infty$, $1/x^2 \rightarrow 0$: result $= 3/1 = 3$.

Answer: 3

7. Evaluate the one-sided limit.

$$\lim_{x \rightarrow 2^-} \frac{x-2}{|x-2|}$$

→ For $x < 2$: $|x-2| = -(x-2)$, so $(x-2)/|x-2| = -1$.

→ Left-hand limit $= -1$.

Answer: -1

8. Determine if the limit exists. Evaluate both one-sided limits.

$$\lim_{x \rightarrow 1^\pm} f(x), \quad f(x) = x^2 \text{ if } x < 1, \quad f(x) = 2x - 1 \text{ if } x \geq 1$$

→ Left-hand limit ($x < 1$): $\lim x^2 = 1$.

→ Right-hand limit ($x \geq 1$): $\lim (2x-1) = 2(1)-1 = 1$.

→ Both sides equal 1, so the limit exists and equals 1.

Answer: $\lim_{x \rightarrow 1^-} x^2 = 1, \quad \lim_{x \rightarrow 1^+} (2x - 1) = 1$

9. Evaluate the limit involving a difference quotient.

$$\lim_{h \rightarrow 0} \frac{(x+h)^2 - x^2}{h}$$

→ Expand $(x+h)^2 = x^2 + 2xh + h^2$.

→ Numerator: $x^2 + 2xh + h^2 - x^2 = 2xh + h^2$.

→ Factor h : $h(2x + h)/h = 2x + h$.

→ As $h \rightarrow 0$: limit $= 2x$.

Answer: $2x$



10. Apply the Squeeze Theorem. Given $-x^2 \leq f(x) \leq x^2$ near $x = 0$, find the limit.

$$\lim_{x \rightarrow 0} f(x) \text{ given } -x^2 \leq f(x) \leq x^2$$

→ $\lim(-x^2) = 0$ and $\lim(x^2) = 0$ as $x \rightarrow 0$.

→ By the Squeeze Theorem: $\lim f(x) = 0$.

Answer: 0

